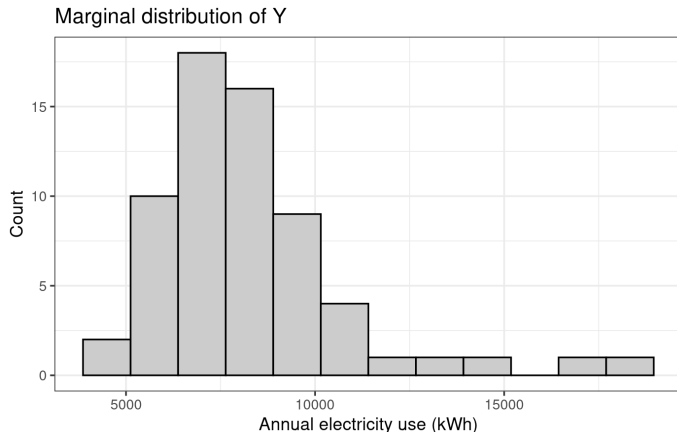


Generalized Linear Models

Grinnell College

October 2, 2026

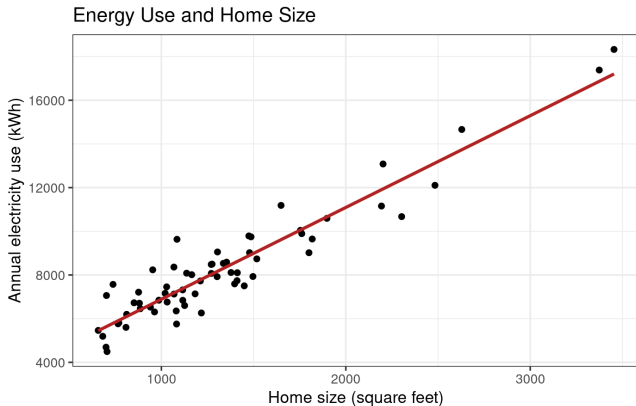
Does the response have to be normal?



Annual electricity use is clearly right-skewed.

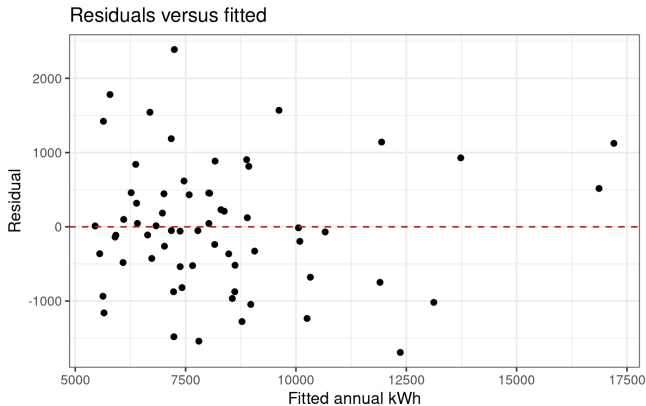
Does this rule out a normal linear regression model?

The model describes the response given the predictors

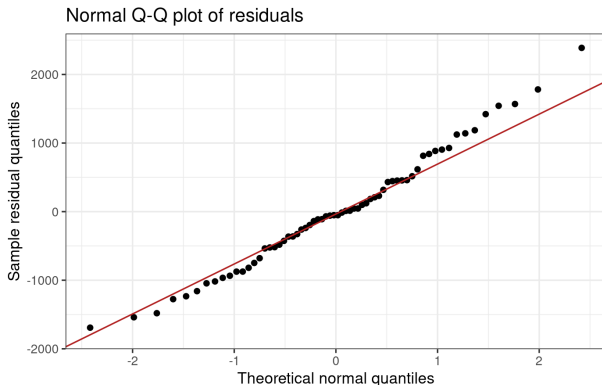


At each fixed home size, electricity use is normally distributed.

Check the conditional errors using residuals



From normal regression to the GLM random component



The residual Q-Q plot is approximately linear.

Normal regression: normality concerns $Y \mid X$, not marginal Y .

GLMs: the random component specifies the distribution of $Y \mid X$, such as normal, binomial, or Poisson.

Snoring and Heart Disease

snoring	yes	no	x
never	24	1355	0
occasional	35	603	2
nearly_every_night	21	192	4
every_night	30	224	5

Linear and Logistic

